Transfer task: mission plan

Organize the assigned evidence. Your plan, your work.

21 A

STEP 1 · SKETCH THE SITUATION

Mark the reference point, current location, motion arrow, and east/north parts.

STEP 2 · GIVENS

QUANTITY	VALUE	UNIT

STEP 3 · MODEL ASSUMPTION

☐ steady velocity ☐ constant acceleration because ____

STEP 4 · FIRST CALCULATION (quantity → numbers + units → result)

STEP 5 · CHECK

☐ direction matches my sketch

☐ size is reasonable

☐ I added the starting position

☐ units on every number

NEXT STEP

I have completed ____, supported by ____. My next step is ____ because ____.

TRANSFER 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Transfer-task planning board

21 D

DIRECTIONS · Complete any three in a row to build your plan. These support — not replace — the assigned task.

DRAW Sketch the situation with a reference point and axes.	**CALCULATE** List every given value with its unit.	**EXPLAIN** Which model fits: steady v or constant a? Why?
REAL WORLD What does the range in the task mean for a decision?	**CHALLENGE ★** Write the full chain of equations you'll need, in order.	**CALCULATE** Do your first calculation, one quantity per line.
EXPLAIN What does the model leave out?	**DRAW** Split the velocity arrow into east and north parts.	**REAL WORLD** Who would use your prediction, and what do they need from it?

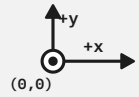
Three in a row: _____ · The square that taught me most: _____

TRANSFER 1 2 3

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Picture dictionary: transfer task

Language goal: I can **plan** a solution.



⊙ (0,0)

reference point

punto de referencia · ponto de referência

where we measure from

my language: _____

use it: _____



v = 8,500 m/s

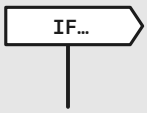
given

dato · dado

a value we know

my language: _____

use it: _____



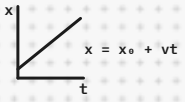
assumption

suposición · suposição

what we pretend is true

my language: _____

use it: _____



x = x₀ + vt

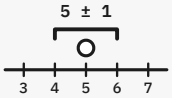
model

modelo · modelo

the math we use

my language: _____

use it: _____



5 ± 1

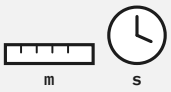
range

rango · intervalo

lowest to highest

my language: _____

use it: _____



m · s · m/s

units

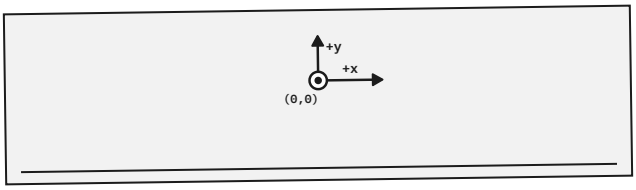
unidades · unidades

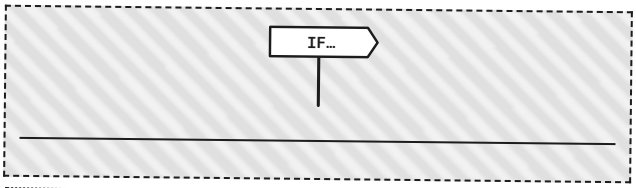
what each number measures

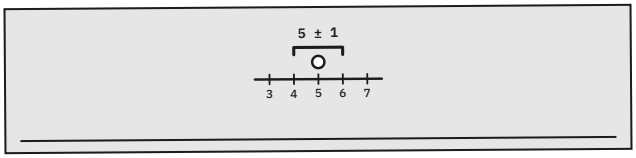
my language: _____

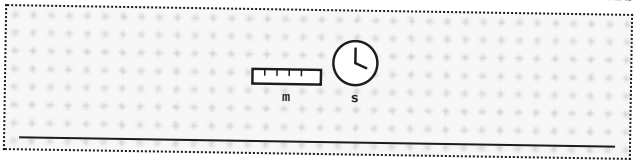
use it: _____

MATCH · WRITE THE WORD FOR EACH PICTURE









I can point and say all 6 words to a partner ☐

TRANSFER 1 2 3

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Say it, then write it: make a plan

Language goal: I can **explain** my plan.



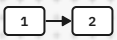
reference point
punto de referencia · ponto de referência
the zero we measure from



given
dato · dado
a number we are told



assumption
suposición · suposição
what we treat as true



step
paso · passo
one part of the plan

English · Spanish · Portuguese. My home language: _____

1 · READ AND FILL · WORD BANK: reference point · given · assume · first · then

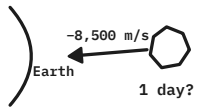
I mark my ____ as zero. I list each ____ with its units. I name what I ____, like steady velocity. ____ I find the change. ____ I add the starting position.

2 · SAY IT TO A PARTNER, THEN WRITE IT



My reference point is ____ and + points ____ .
I assume ____ because ____ .

3 · SOLVE WITH THE STEPS



2026-XJ is 1.2 × 10⁹ m from Earth and moves 8,500 m/s toward Earth. Plan how to find where it is after 1 day.

STEP 1 · UNITS
1 day = ____ s

STEP 2 · CHANGE
Δx = ____ × ____ = ____ m

STEP 3 · SAY IT
"After 1 day it is ____ m from Earth."

4 · EXPLAIN IT · CHOOSE A FRAME

☐ My first step is ____ because ____ .
☐ I assume ____ ; if that is wrong, ____ .
☐ First ____, then ____, so ____ .

I said it ☐ · I wrote it ☐

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VOL. 1 · NO. 21 · TRANSFER REPORTER: _____

The 2026-XJ Daily

Press release draft: what we know so far

HEADLINE (no final answers yet — what is the plan?)

THE STORY

Describe the situation, the evidence you were given, and the model you will use.

WHAT WE WILL MEASURE / CALCULATE

1. ____
2. ____
3. ____
4. ____

SKETCH FOR READERS

LIMITS OF THE MODEL

What could make our prediction wrong?

EDITOR'S NOTE TO MYSELF

Tomorrow I still need to ____ .

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Plan it with the crew

21 R

DIRECTIONS · The crew is starting the transfer task. Reply with planning moves — not answers.

XJ-7 transfer team

ELI

ok where do we even start

YOU ① first move

MAZ

we have like 6 numbers. do we just multiply them?

YOU ② sort the givens

NADIA

steady velocity or constant acceleration?

YOU ③ choose a model

ELI

my answer is negative. did we mess up?

YOU ④ check it

SHOW YOUR RECEIPTS

① sketch + reference point

② quantity · value · unit

③ my assumption and why

④ direction, size, start

TRANSFER 1 2 3

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Comic: brief the new intern

21 CS

YOUR READER

A new intern on the 2026-XJ team who just arrived today.

Stick figures are fine. The physics gets checked.

MUST-HAVES · check it off, then color its circle

- ☐ ◌ **BLUE** — A measured position with a unit
- ☐ ◌ **PURPLE** — A velocity arrow or a velocity value
- ☐ ◯ **PENCIL** — The equation, written out in panel 2

1 · WHAT WE MEASURE *Show the team taking a measurement.*

CAPTION _____

2 · WHAT WE CALCULATE *Show which equation they use, and why.*

CAPTION _____

3 · WHAT WE PREDICT *Show the prediction and how sure they are.*

CAPTION _____

EXPLAIN IT · In 1–2 sentences, what physics does your comic show? Use: **measure · equation · predict · uncertainty**

TRANSFER 1 2 3

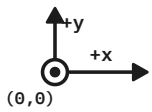
UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Doodle Notes · reference frame

I can draw a change and explain what my marks mean.

Name: _____ My home language: _____

Reference frame · sistema de referencia (ES) · referencial (PT)



TODAY'S CASE

A team reports that fictional asteroid XJ is "400 km away." They have not said away from what.

01 · SKETCH AND LABEL

Draw Earth, XJ and a blue position axis. Label the zero, the positive direction and units.

02 · CHANGE THE PICTURE

Add a second measured position at a later time. Mark what was measured and which assumption a forecast would need.

Annotate your sketches. Circle the mark that best shows reference frame. Add a label in English or your home language.

SAY IT TO A PARTNER · THEN WRITE

A second team can interpret my position because …

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced
☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else

Frayer Model · reference frame

I can explain a term by comparing an example with a nonexample.

Name: _____ My home language: _____

Reference frame · sistema de referencia (ES) · referencial (PT)

A · "It moved forward" specifies a frame without naming a reference.

B · A position is reported from a chosen zero along a named positive axis.

01 · DEFINITION

What does reference frame mean in your own words? Use the notes or start in your home language.

02 · CHARACTERISTICS

What must be true? Label a small sketch to show one feature.

03 · EXAMPLE

Choose A or B as your example. Sketch it and explain which feature makes it fit.

04 · NONEXAMPLE

Explain why the other card does not fit. Change one detail so that it would.

SAY IT TO A PARTNER · THEN WRITE

I chose card ___ as my example because …

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